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Low-Temperature Cross-Linking of PEDOT:PSS Films Using Divinylsulfone

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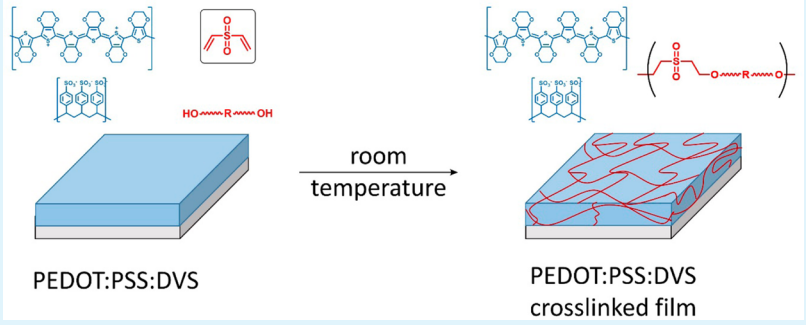
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Supporting Information



ABSTRACT: Recent interest in bioelectronics has prompted the exploration of properties of conducting polymer films at the interface with biological milieus. Poly(3,4-ethylenedioxythiophene) doped with poly(styrenesulfonate) (PEDOT:PSS) from a commercially available source has been used as a model system for these studies. Different cross-linking schemes have been used to stabilize films of this material against delamination and redispersion, but the cost is a decrease in the electrical conductivity and/or additional heat treatment. Here we introduce divinylsulfone (DVS) as a new cross-linker for PEDOT:PSS. Thanks to the higher reactiveness of the vinyl groups of DVS, the cross-linking can be performed at room temperature. In addition, DVS does not reduce electronic conductivity of PEDOT:PSS but rather increases it by acting as a secondary dopant. Cell culture studies show that PEDOT:PSS:DVS films are cytocompatible and support neuroregeneration. As an example, we showed that this material improved the transconductance value and stability of an organic electrochemical transistor (OECT) device. These results open the way for the utilization of DVS as an effective cross-linker for PEDOT:PSS in bioelectronics applications.

KEYWORDS: *divinylsulfone, PEDOT, cross-link, conducting polymers, biocompatible*

1. INTRODUCTION

Bioelectronics is currently a fast growing field driven by many new discoveries by a dedicated scientific community.[1,2] Prominent topics are the creation and design of implantable electronic medical devices (IEMDs) that require long-term stability, such as cochlear implants, retinal prostheses, microfabricated cortical electrodes, cutaneous electrodes, and more. In literature, polythiophenes, in particular poly(3,4-ethylenedioxythiophene (PEDOT), are considered to be the main protagonist in these applications. PEDOT is mainly doped with an anionic polymer poly(styrenesulfonate) (PSS) and is commercially available on the market as a blue dispersion in liter scale, bearing the name Clevios. The conductivity of the pristine film that could result from spin-coating of Clevios, ranges between 0.1 and 1 S cm^{-1}. In order to be processable and create films for application in most organic electronic devices, it is necessary to elevate this value. Many studies have been carried out in this race, leading to conductivity values up to 3000 S cm^{-1}.[3]

Biocompatibility is another key point in designing new bioelectronics devices.[4] Many studies are presented in literature demonstrating that films of PEDOT:PSS are stable and compatible *in vivo* and *in vitro*.[5−8] Even though many of these strategies present a stable material in contact with a living tissue, in a long-term point of view, cross-linking is needed to prevent delamination and eventual redispersion of the

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conducting film, especially on glass substrate.[9,10] In order to achieve this, numerous methods were used such as UV light or poly(ethylene oxide) treatment.[11−13] Among them, one of the most studied is the use of silanes,[14,15] in particular glycidoxy propyltrimethoxysilane (GOPS).[7,16−20] In spite of the excellent ability to cross-link PEDOT:PSS film, GOPS presented some important drawbacks such as its high curing temperature (140 °C for 1 h) and negative effect on the conductivity values. Due to the natural insulating properties of the siloxane network, this cross-linker is normally used together with doping agents such as ethylene glycol or DMSO to reduce the loss of conductivity generated by GOPS.[21] Its main inconvenience is the high curing temperature needed (>100 °C) which prevents applying the process in the presence of any biological content. In our search for a better candidate to cross-link PEDOT:PSS, we explored the possibility of using divinylsulfone (DVS), which is a broadly studied cross-linker for biomolecules. DVS is a reactive molecule against nucleophiles, with two active sites, in correspondence to the outer carbon in the double bond.[22] It has been fruitfully applied many times to cross-link biomolecules such as glycosaminoglycanes (GAGs), for example, hyaluronic acid, and has already been implemented in many products for *in vivo* applications.[23] The advantages of DVS are its high reactivity, possibility to work in an aqueous environment, and low boiling point that facilitate the elimination of excess.

Conducting polymer-based devices are widely used in the bioelectronics field for electrophysiology and *in vivo* and *in vitro* biosensing applications. One promising candidate standing from this field is the organic electrochemical transistor (OECT) as a transducer of biological events with local amplification such as the movement of biomolecules, electroencephalography activity, probe cell adhesion and more.[24−26] In OECTs, the conducting polymer film between drain and source contacts is placed in directly in the electrolyte. Upon application of a bias at an external gate that is immersed in the electrolyte, ions are injected in the bulk of the polymer. This subsequently alters the hole density of the film, thus the current flowing between drain and source electrodes. Therefore, OECTs act as an effective ion-electron transducers which are governed by the channel materials ability to conduct holes as well as ions. This phenomenon can be analyzed using transconductance as a figure of merit, which determines the conversion between gate voltage (V_G) and drain current (I_D). A small increase in transconductance means a much higher recording capacity of small biological signals. The reliability and stability of an OECT is determined by the stability of this parameter, which becomes especially relevant in the case of long-term *in vivo* recordings.[27]

In this work, we applied DVS as an alternative cross-linker of PEDOT:PSS to enhance the stability for long-term applications and allow the resultant thin films of the suspension to be cured at ambient temperature. The suspension of cross-linked material was characterized chemically through NIR−UV−vis, Raman, and FT-IR. For physical assessment of cross-linked films, conductivity measurement showed that DVS-cross-linked PEDOT:PSS showed conductivity enhanced by 2-fold compared to that of GOPS-cross-linked PEDOT:PSS. Although biocompatibility of PEDOT:PSS has been widely described in literature,[28] we confirmed biocompatibility of DVS-cross-linked PEDOT:PSS by cytotoxicity testing according to ISO 10993−5 guidelines, and additional biocompatibility tests were performed using the neuroblastoma cell line SH-SY5Y. We further tested the PEDOT:PSS:DVS formulation in organic electrochemical transistor (OECT configuration) by employing thin films of PEDOT:PSS:DVS as the channel material. We investigated the transconductance as a function of time for subsequent comparison of performance and stability of it to traditional PEDOT:PSS channels.

2. EXPERIMENTAL SECTION

2.1. Materials and Method. PEDOT:PSS Clevios PH1000 was bought from Heraeus. Deuterated solvent and NMR tubes were bought from Deutero GmbH. 3-(Trimethoxysilyl)propyl methacrylate and soap micro-90 were supplied by Sigma-Aldrich. All other chemicals were supplied from Fisher Scientific. Microscope glass slides with cut edges were provided by VWR (76 × 26 × 1 mm^3). Round-shaped glass coverslips were bought from OPPAC S.A. in two different sizes: 1.3 and 2.5 cm diameter. Fourier-transform Raman spectroscopy was performed on a Bruker RFS 100/S with YAG:Nd laser (1064 nm) 500 mW horizontally polarized (H) as excitation source, ROCKSOLID configuration permanently aligned as interferometer, a HeNe laser at 633 nm and 1 mW as alignment excitation source, and Ge-diode as detector, with cooling at liquid nitrogen temperature (77 K). A 5 mm mirror quartz cuvette was filled with a suspension of PEDOT:PSS:DVS diluted 1:50, running 150 scans per measure. The UV−vis−NIR measurements were done using a PerkinElmer Lambda 950 between 200 and 1350 nm using quartz cuvette and preparing PEDOT:PSS:DVS suspension diluted 1:10. ^{1}H NMR has been performed on a Bruker Avance 400 MHz, using WATERGATE and D_2O. FT-IR spectra were recorded on a Bruker Alfa ATR.

2.2. PEDOT:PSS Suspensions. *2.2.1. PEDOT:PSS:GOPS Suspension.* To prepare 25 mL of aqueous dispersion, 5 mL of ethylene glycol, 25 μL of dodecyl benzenesulfonic acid (DBSA), and 250 μL of (3-glycidyloxypropyl) trimethoxysilane (GOPS) were mixed with 20 mL of PEDOT:PSS Clevios PH1000.

2.2.2. PEDOT:PSS:DVS Suspension for Conductivity Measurements. To prepare about 25 mL of aqueous dispersion, 1.25 mL of ethylene glycol and 25 μL of dodecyl benzenesulfonic acid (DBSA) were mixed with 20 mL of PEDOT:PSS Clevios PH1000;to this suspension,different quantities of DVS were added. Once prepared, it is preferable to use the dispersion immediately or in the next hour.

2.2.3. PEDOT:PSS:DVS Suspension for Devices. The procedure is the same as that for PEDOT:PSS:DVS, but 50 μL of (3-glycidyloxypropyl) trimethoxysilane (GOPS) was mixed in the formulation.

2.3. Films Preparation for Biotests. Round-shaped glass coverslips in two different sizes, 1.3 and 2.5 cm diameter, were used. Piranha solution formed by mixing one part of H_2O_2 and one part of H_2SO_4 was used to pretreat the coverslips. Spin-coated films were made by direct solution dropping on a KLM SCC spin-coater and spin-coating them for 40 s at 1000 rpm (to get a film of about 100 nm). A suspension of 3% GOPS (v/v) was used together with two solutions containing 5 and 10% (v/v) DVS, prepared as described before. Once spin-coated, glass slides were hard-baked for 1 h at 50 °C under vacuum. To prove delamination, a dry sample coverslip was left standing in a beaker with 20 mL of 0.01 M phosphate-buffered saline (PBS), changing this solution 3 times every 8 h.

2.4. Film Preparation for Conductivity Measurements. Prior to utilization, glass slides were sonicated in a 2 wt % soap bath during 10 min, followed by sonication in an isopropanol:acetone 2:1 bath during 10 min. After sonication, the glass-slides were rinsed with isopropanol and DI water successively. Afterward, they were activated by plasma treatment at 100 W with oxygen at 40 sccm for 30 s on a E-100 Benchtop Plasma System by PlasmaEtch, NV, USA. The solutions were spin-coated on a Laurell WS-400B-6NPP-LITE at 1000 rpm for 40 s for the ones containing DVS. For the suspension without DVS, the spin-coating was 30 s at 3000 rpm, resulting in both cases in films about 100 nm thick. Bruker Dektak profilometer was used to measure film thickness. Once spin-coated, glass slides were hard-baked for 40 min at 140 °C. The measurements of conductivity were performed on a 4-point probe method, using a Jandel KM3-AR instrument.

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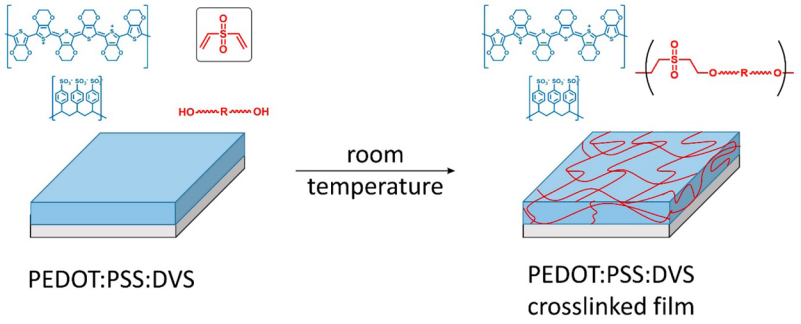


Figure 1. Schematic representation of cross-linking reaction of PEDOT:PSS film by DVS.

2.5. OECT Fabrication. Glass slides (76 × 26 × 1 mm^3) were cleaned by sonication in a 2 wt % soap solution in DI water for 15 min and then in isopropanol/acetone 1:1 v/v for 15 min. Au interconnects were patterned by spin-coating S1813 (Shipley) photoresist on the glass slide, exposing it to UV light using a SUSS MJB4 contact aligner, and developed using MF-26. A 10 nm Cr adhesion layer, followed by a 100 nm Au film, were deposited (Alliance Concept EVA450) and patterned using lift-off in acetone. Afterward, an insulation layer 1.7 μm of parylene-C with 3-(trimethoxysilyl)propyl methacrylate as adhesion promoter was deposited. A layer of soap (micro-90) and an extra sacrificial layer of parylene-C (2 μm) were deposited. The last fabrication step was the patterning of a 3.5 μm layer of photoresist, AZ9260, which was then etched using oxygen plasma using an Oxford 80 plus. In this way, a channel area was defined, 100 μm width and 10 μm length. The transistor was spin-coated with PEDOT:PSS containing 5% v/v ethylene glycol, 0.1% v/v dodecyl benzenesulfonic acid, 3% v/v divinyl sulfone, and 0.2% v/v (3-glycidyloxypropyl) trimethoxysilane at 1000 rpm for 40 s, resulting in a film ∼100 nm thick. After peeling off the upper layer of parylene, the PEDOT:PSS:DVS channel was put in direct contact with an electrolyte solution (0.1 M NaCl) where the Ag/AgCl gate electrode (Warner Instruments) was submerged for measurements. Drain and gate voltage and current were measured and applied correspondingly by a Keithley 2602A instrument.

2.6. Biocompatibility Testing L929 and SH-SY5Y Cell Culture. Murine L929 cells were cultured in in high glucose Dulbecco's modified Eagle's medium (DMEM; Sigma, ref. D5796) supplemented with 10% fetal bovine serum (FBS; Biochrom, ref. A1020S0115) and 1% penicillin−streptomycin (Pen/Strep; Lonza, ref. A3717−745E). Cells were passaged at ∼80% confluence or seeded according to experimental preferences. Human SH-SY5Y cells were cultured in Advanced D-MEM/F12 (Gibco, ref:12634−010) supplemented with 10% FBS and 1% Pen/Strep, and cells were passaged or seeded for experiments at ∼80% of confluence. Both cell lines were maintained in a humidified incubator at 37 °C and 5% CO_2.

2.7. Cytotoxicity Assay According to ISO 10993−5 Guidelines. Cytotoxicity or biocompatibility of samples was assessed by MTT (Roche, Cell proliferation kit I) assay performed following the ISO 10993−5 guidelines for the extract method using the murine fibroblast cell line L929. The protocol for MTT was performed according to manufacturer instructions, and all cultures and incubation steps were performed in a humidified incubator at 37 °C and 5% CO_2. Cells were seeded in a 96-well plate at a density of 10^4 cells/well. Extracts of samples or negative (high-density polyethylene, USP, ref. 1546707) and positive controls (positive bioreaction, USP, ref. 1071439) were generated by conditioning cell culture medium for 24 h with samples of appropriate dimensions defined in ISO 10993−5. This standard states that samples with a thickness below 0.5 mm should have at least a contact surface area of 6 cm^2. If the thickness of samples is between 0.5 and 1 mm, then this surface area has to be at least 3 cm^2; with a sample thickness of 1 mm or greater, this value must be at least 1.25 cm^2. At 24 h after seeding, L929 cells were cultured in these medium extracts for 24 h abd; thereafter, MTT was added for 4 h. To dissolve purple formazan salt crystals, cell proliferation kit I solubilization solution was added for 24 h, and absorbance was measured at wavelength 550 nm using a Multiskan Ascent (Thermo Scientific) plate reader. Outcomes were interpreted by using unpaired Student's t test; differences which showed $p < 0.05$ were considered as significant.

2.8. Proliferation of SH-SY5Y Cells. Proliferation of SH-SY5Y cells on the material substrates was evaluated by counting of live cells. SH-SY5Y cells were seeded in 24-well plates at a density of 4 × 10^4 cells/well in Advanced D-MEM/F12 (Gibco, ref: 12634−010) supplemented with 10% FBS and 1% Pen/Strep. Cells cultured on tissue culture plastic (TCP) were used as a control condition. A 24 h after seeding, cell culture medium was refreshed, or live/dead assay was started at the t = 0 h time point. Live/dead assay (Thermofisher, ref: L3224) was performed according to manufacturer protocol. Cells were washed once with serum-free culture medium, and then cells were treated for 15 min with the provided 8 μM EthD-1 for staining of dead cells and 4 μM Calcein-AM for staining of live cells. Imaging was done with the use of an Olympus IX51 fluorescent microscope. Proliferation of SH-SY5Y cells was evaluated by counting the amount of live cells in a minimum of 3 microscopic fields at 10× in 3 independent samples per time point (0, 24, and 48 h). Counting of cells was done by using the cell counter plugin of the ImageJ software.

2.9. Neuronal Differentiation of SH-SY5Y Cells and Immunocytochemistry. For differentiation of SH-SY5Y cells, cells were seeded at a density of 6 × 10^4 cells on laminin-coated (10 μg/mL) glass coverslips (control) or PEDOT:PSS:DVS films drop-cast on glass coverslips. Differentiation of SH-SY5Y cells was performed as described previously with some modifications.[29] Induction of neuronal differentiation of SH-SY5Y cells was done by sequential treatment of cells with 1 μM synthetic retinoid ec23 (Amsbio, ref. SPR002) for 3 days, followed by 3 days of treatment with human brain derived neurotrophic factor (BDNF; Sigma-Aldrich, ref. B3795). Both steps were performed in serum free medium. For analysis of neurite formation, immunocytochemistry was performed. After rinsing with PBS 1× cells were fixed in 4% formaldehyde in PBS 1× (0.1% Triton X-100) for 12 min at 4 °C followed by rinsing twice with PBS 1× for 10 min and 1 h of blocking in 1% BSA in PBS 1× at room temperature. Cells were stained with anti-neurofilament 200 primary antibody (1:500; Sigma-Aldrich, ref. N4142) for 1.5 h at room temperature Thereafter, cells were rinsed 2 times with PBS 1× and incubated (1:500) with Alexa 568 goat anti-Rb IgG (H+L; Invitrogen, ref. A11011) for 1.5 h at room temperature. After incubation with secondary antibody, cells were washed with PBS 1× supplemented for nuclear staining with Hoechst (1:1000), followed by 5 min of PBS rinsing. Permafluor (Thermo scientific, ref. TA-006-FM) was used to mount the coverslips up to the glass slides for fluorescence imaging. ImageJ was used to measure the length of formed neurites together with the simple neurite tracer plugin, and results were compared with unpaired Student's. Results where $p < 0.05$ were taken as significant.

3. RESULTS AND DISCUSSION

The chemical structure of DVS is shown in Figure 1. Different amounts (1−40 by %v/v) were added to a commercial PEDOT:PSS solution (Clevios PH1000), and films were prepared by casting or spin-coating. After curing at room

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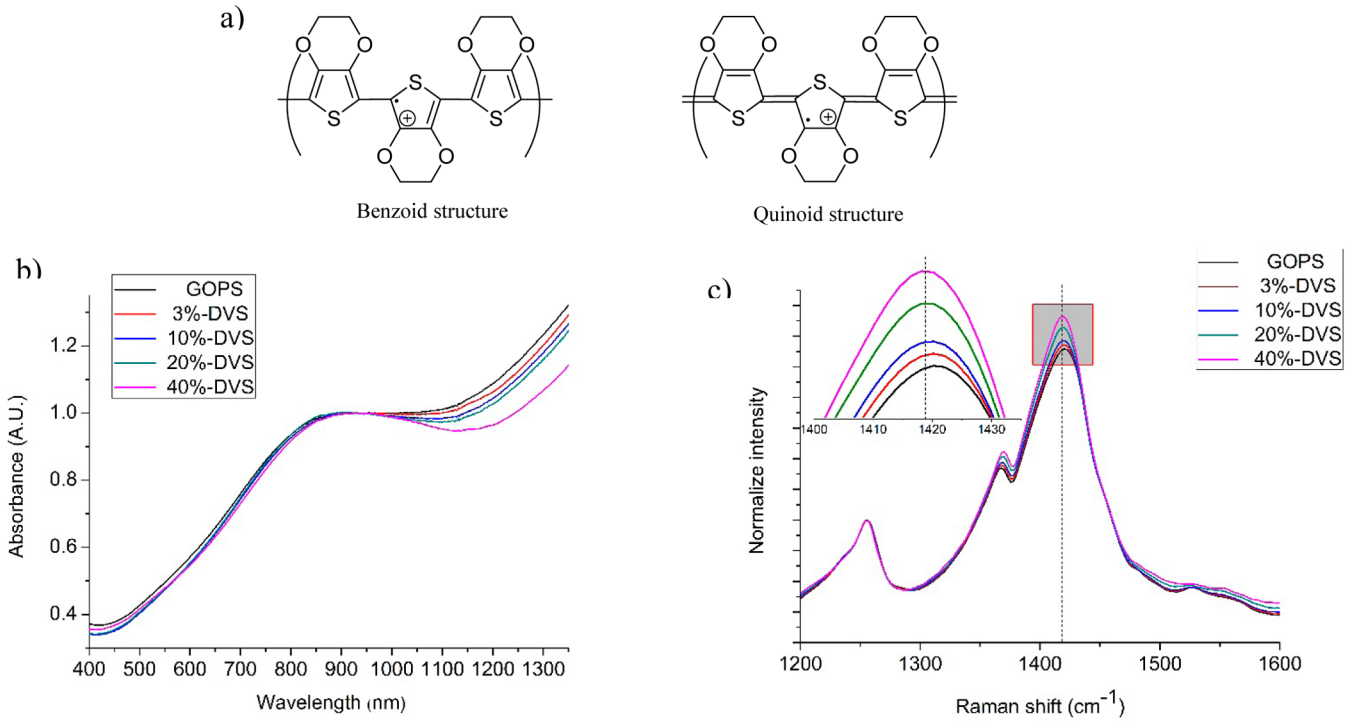


Figure 2. (a) Resonance structure of PEDOT chain in "benzoid" and "quinoid" form, (b) UV−vis−NIR, and (c) Raman spectrum of PEDOT:PSS:DVS at different concentrations of DVS.

temperature for 14 h, the films were cross-linked. Raising the temperature to 50 °C speeds up the cross-linking process to only 1 h. In order to prove the cross-linking, the films were immersed in water or PBS. As it can be seen in Figure S1, a not-cross-linked film is going to delaminate in less than a minute, while the films cross-linked using DVS are stable and do not disperse or delaminate.[10]

3.1. Cross-Linking Mechanisms of PEDOT:PSS:DVS. In order to investigate in detail the chemical nature of the relevant cross-linking mechanism, we first looked at the dispersions. The easy preparation of dispersions for characterization consisted only in mixing the desired amount of DVS with the commercial aqueous suspension of PEDOT:PSS. The solution was sonicated to homogenize the suspension and left overnight in a closed vial, to avoid the evaporation of DVS. UV−vis−NIR spectra (Figure 2b) show only one strong band between 800 and 1100 nm arising from the polarons present in the PEDOT chains.[30] With increasing amounts of DVS, a decrease in the intensity at higher wavelengths can be seen. As reported by Massonet and co-workers in 2014, this reduction can be interpreted as a decrease in the quantity of bipolarons and increase of the radical species of the conjugated thiophene backbone.[31]

This confirms that the material is highly doped and that the addition of DVS does not deplete it. In the Raman spectroscopy, shown in Figure 2c, the addition of DVS does not change drastically the aspect of the spectra as we can see from the comparison between the black line and the others. Analyzing the spectra, we can notice how the bands at 1433 and 1260 cm^{-1} corresponding to symmetrical $C_\alpha-C_{\alpha'}$ inter-ring stretching vibrations, remain unchanged.[32−34] The highest band located at 1420 cm^{-1} and the one at 1522 cm^{-1} also remain barely unchanged in size but upon analyzing closely are red-shifted. These two bands correspond to symmetrical and unsymmetrical stretching vibrations of $C_\alpha=C_\beta$. The redshift indicates that the structure of the carbon backbone is turning more into the quinoid form as is shown in Figure 2a.[35,36] Another confirmation of this phenomena is the blueshift of the band at 1360 cm^{-1}. This band corresponds to the stretching deformations of aromatic $C_\beta=C_\beta$: in this case our structure, tending to the quinoid structure, is losing aromaticity and bond order is decreasing.[37] The small increase in both bands is possibly due to the formation of hydroxide anions, as pointed out in the work of Cruz-Cruz et al.[38] In fact, adding DBU (1,5-diazabiciclo(5.4.0)undec-5-ene), they noticed the formation of hydroxide anions that quench the polarons, increasing the intensity of the Raman shifts. This is due to strong resonant effects of the neutral form of thiophene.

The same results can be observed from the FTIR (Figure S2) spectra conducted on dry films. As affirmed in literature, a growth of the bands between 1600 and 100 cm^{-1} is corresponding of a more doped material, affordable with solvent treatment as shown by Nguyen et al.[39,40] To confirm the presence of the sulfone moiety, bands at 1360 and 1152 cm^{-1} are strong evidence, but unluckily only in the last spectrum, when the concentration of DVS is 20%, those bands overtake the rest and become visible.

3.2. Film Formation, Electrical Properties, and Device Performance. *3.2.1. Film Formation and Conductivity Measurements of PEDOT:PSS:DVS Films.* Next, we investigated the formation of PEDOT:PSS:DVS films obtained by curing the dispersions at 50 °C for 1 h. Unfortunately, comparison of the FTIR and NMR spectra of the DVS-treated and those of the nontreated films did not show any clear difference. In order to explain the cross-linking nature, we synthesized a model suspension of PEDOT:PSS, respecting the same ratio and solid content as declared by the producer, using different oxidant agents (Supporting Information, section 3). With this model suspension, we were not able to obtain an insoluble film after curing with DVS, but when we add ethylene

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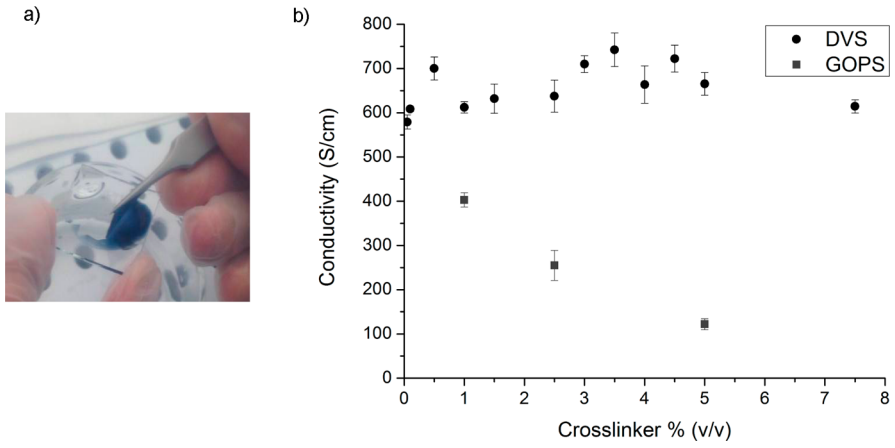


Figure 3. (a) Lifting and manipulation of PEDOT:PSS:DVS from a glass substrate. (b) Conductivity measurements of films prepared with solution containing different amount of DVS or GOPS cross-linker.

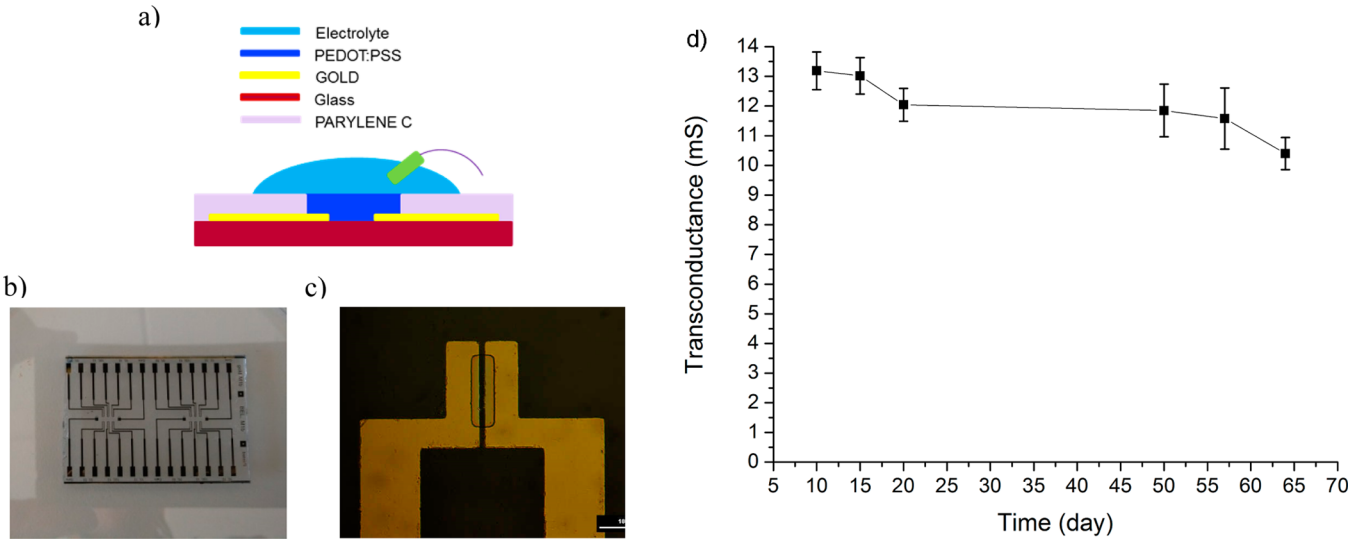


Figure 4. (a) OECTs design, (b) OECT device, (c) OECT PEDOT:PSS:DVS channel 100 μm × 10 μm, and (d) transconductance evolution with time for channel formulation PEDOT:PSS:DVS with channel size 100 μm*10 μm.

glycol (20%), the result is a cross-linked film. This is because DVS reacts very fast with nucleophiles such as alcohols, thiols, and amines.[22] It is known that Clevios PH1000 is a product of a spread range of suspensions, of which the complete and detailed composition is unknown but includes stabilizers, viscosity additives, dopant improvers, and conductivity enhancers such as ethylene glycol.[41,42] The model reaction between DVS and ethylene glycol has been tested and followed by ^{1}H NMR, leaving DVS and ethylene glycol in water at the same pH of the suspension of PEDOT:PSS (pH ∼2) (Figure S3). The analysis confirms the reaction between DVS and the diol, shielding the singlet of the methylenes of the diol and eliminating the signals at high frequency of the vinyl moiety. All-in-all, we can conclude that the chemical nature of its cross-linking by reacting with some existing nucleophiles (alcohols and amines) in the PEDOT:PSS formulation. Next, we investigated the effect of the DVS cross-linking into the conductivity of the PEDOT:PSS films. To measure film conductivity, the 4-point probe (4PP) method was used.[43,44] For this study, the conductivity of several PEDOT:PSS:DVS formulations were measured and compared with reported values for the widely applied PEDOT:PSS:GOPS cross-linked films (Figure 3b). PEDOT films cross-linked with DVS present conductivity in between 550 and 750 S cm^{-1} which is 200−300 S cm^{-1} higher than those of GOPS, at any given cross-linker concentration. Measurement fluctuations are explained due to the soft nature of PEDOT films which difficult the precision of the measurements obtained by the 4PP technique, phenomena already observed in a previous study.[43] Interestingly, the conductivity values stay constant with the DVS content, whereas in the case of GOPS a fast decrease in conductivity is observed when the DVS concentration increases above 1%. With these results, we can conclude that DVS is a more appropriate PEDOT cross-linker than GOPS, as it does not reduce PEDOT film conductivity. In fact, the low boiling point of DVS, 25 °C at 2 mm Hg, and the high reactivity of the divinyl scaffold make it react totally, with the excess being eliminated during the baking treatment at 50 °C under vacuum; it is good to point out that as DVS is highly volatile the use of vacuum could be avoided if the baking process is done at higher temperature as normally used in OECT preparations (above 140 °C). However, this is not occurring when GOPS is used as cross-linker because it has a higher boiling point (120 °C at 2 mm Hg) and because siloxanes are less reactive groups.

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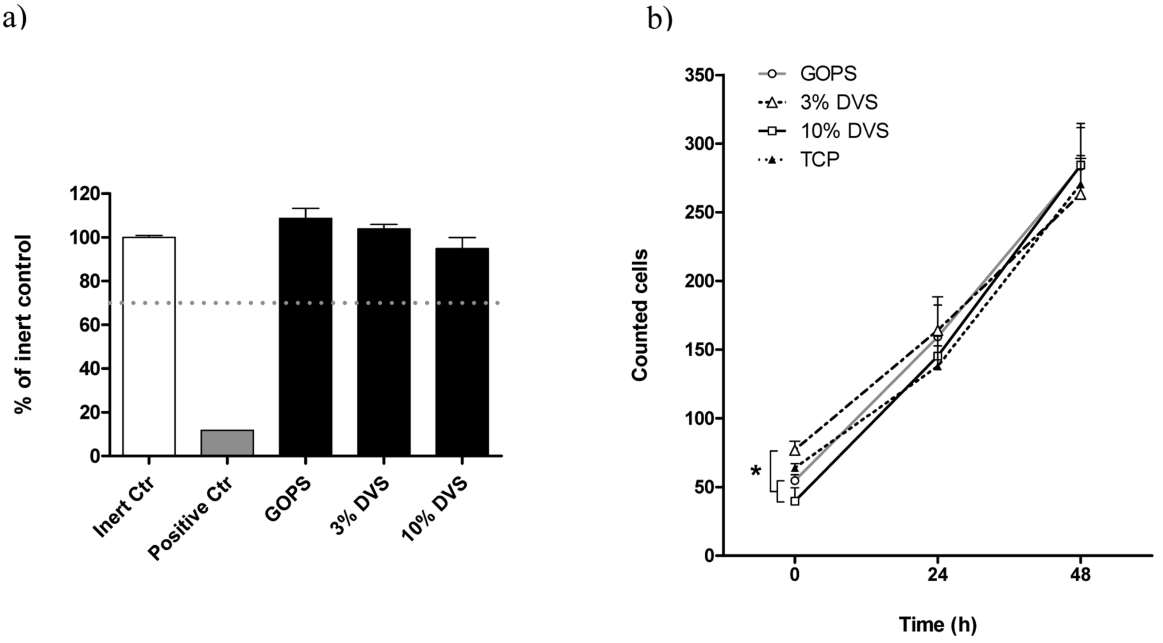


Figure 5. (a) MTT cytotoxicity assay was performed according to ISO 10993−5 guidelines for extracts, using L929 cells and medium extracts of PEDOT cross-linked with 3% GOPS, 5% DVS, or 10% DVS (v/v). (b) Proliferation assay of SH-SY5Y cells cultured on tissue culture plastic (TCP) or PEDOT cross-linked with GOPS, 5% DVS, or 10% DVS during 48 h (t = 0 h represents 24 h after seeding). After live/dead assay, analysis was performed by counting live cells. Statistical analysis performed with unpaired Student's t test with * denoting $p < 0.05$.

Moreover, it has been reported that adding solvents such as DMSO or additives such as ethylene glycol or ionic liquids inside the formulation improves the electrical conductivity. In our case, DVS acts as a conductivity enhancer additive, and the cross-linked films show very high conductivity values.[21,45,46]

3.2.2. OECT Transduction Measurements. We further tested ion and hole transport capabilities of the PEDOT:PSS:DVS in an OECT configuration (Figure 4a).[24] Among all measurable parameters in these devices, transconductance is the most important parameter as it indicates its amplification capacity;[26] in other words, it quantifies its efficiency of signal transduction. Transconductance is defined as $g_m = \Delta I_D/\Delta V_G$, where I_D is the drain current and V_G is the gate voltage. In most OECTs present in literature, the channel is formed by the conducting polymer PEDOT:PSS,[47−49] which is previously treated with cross-linkers to avoid its dissolution.[50−52] In this study, we measure transconductance of an OECT, the channel of which is formed by a thin film of PEDOT:PSS:DVS. PEDOT formulation with 3% DVS was chosen for the device fabrication because it ensures stable films with high conductivity. To the latter, 0.2% GOPS was included in the formulation to provide attachment to the substrate as DVS only provides fixation but not attachment. Conductivity suffered almost no decrease with GOPS addition at this concentration (623 ± 19 S cm^{-1}, n = 4). In Figure 4c, transconductance along time is measured for a channel geometry of 10 × 100 μm and 100 nm thickness. Measurements were taken after OECT conditioning to avoid data fluctuations due to an initial PEDOT layer degradation. The OECT was kept in sterile PBS in between measurements, and after rinsing with Milli-Q water, drops of NaCl 0.1 M solution were deposited on the channel for measurements. As is visible in the graph, at day 10, transconductance has a value of 13.2 ± 0.65 mS (n = 4) which is an outstanding value when compared to studies employing PEDOT:PSS:GOPS channels. For channel sizes of 10 × 100 μm and 400 nm thickness, values of 2−3 mS were obtained.[26] Moreover, PEDOT:PSS:DVS channels seem stable in the long term, showing 6 and 20% decay in transconductance after 20 and 64 days, respectively.

PEDOT:PSS:DVS channel shows outstanding transconductance values and good stability in a wide range of frequency and over time (Figures S8 and S9). This means, with respect to GOPS which also shows good stability over time, a big amplification capacity of the OECT when PEDOT:PSS is cross-linked by DVS than by other cross-linkers.[10,25,53] This difference in transconductance is determined by ion transport through the channel which depends on its structure and swelling degree,[54] implying that PEDOT:PSS:DVS film absorbs more water and induces greater phase segregation and structural modification in between PEDOT and PSS chains.

3.3. Biocompatibility of PEDOT:PSS:DVS Films. In order to assess the biocompatibility of PEDOT:PSS:DVS, cytotoxicity of PEDOT:PSS:GOPS was compared to that of PEDOT:PSS cross-linked with DVS, and all samples were compared to an inert (high-density polyethylene) and positive control (polyvinyl chloride). Cytotoxicity testing was performed by MTT assay according to ISO 10993−5 guidelines using the murine L929 fibroblast cell line. L929 cell culture medium was conditioned with PEDOT:PSS:GOPS and DVS cross-linked films, as well as with the inert and positive controls. L929 cells were cultured thereafter for 24 h in this conditioned medium, and cytotoxicity was determined. As expected the positive control showed high cytotoxicity (~10% cell viability); all other samples showed cell viability comparable to that of the inert control around 100%, as shown in Figure 5a. These results show that the films do not release toxic degradation products during the 24 h incubation period.

3.3.1. Proliferation of SH-SY5Y Neuroblastoma Cells on PEDOT:PSS:DVS Films. Since a large part of IEMDs using PEDOT are destined to be used in the CNS environment, SH-SY5Y cells were chosen to serve as an *in vitro* neuronal model cell line to assess biocompatibility in such an environment. In

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addition to biocompatibility in the sense of cell survival, to be able to perform long-term recordings or stimulations of neural tissue, extensive research has been dedicated to create materials or coatings of electrodes that present conditions for optimal cellular attachment and thus minimize the gap of tissue-electrode interface for optimal performance.[55,56] To assess the effect of PEDOT:PSS:DVS on these factors, live/dead staining was performed on SH-SY5Y cells cultured on films prepared as described in previous sections in 24-well plates. At 24 h after seeding, live and dead cells were stained with Calcein-AM and EthD-1, respectively, and cells were manually counted using the ImageJ cell counter plugin. This procedure was repeated at 24 and 48 h. At 0 h, PEDOT:PSS:DVS (3%) showed a significantly higher amount of live cells compared to those with PEDOT:PSS:GOPS and PEDOT:PSS:DVS (10%). This indicates that of the samples analyzed here cross-linking with 3% DVS provides the best conditions for initial cellular attachment. Regarding proliferation, no significant differences between samples and tissue culture plastic control were found over the 48 h time course of the proliferation experiments (Figure 5b). These results show that PEDOT:PSS cross-linked with 3 and 10% DVS supports cellular attachment and proliferation, with 3% being preferred due to the initial difference in cellular attachment at 0 h.

3.3.2. Neurite Length of Differentiated SH-SY5Y Cells on PEDOT:PSS Substrates. When IEMDs are implanted in a CNS environment, it is important that these biomaterials support and possibly stimulate neuronal regeneration. A general method used to assess whether or not a biomaterial supports neurobiocompatibility and neuroregeneration is the measurement of neurite lengths of newly formed neurites in differentiated neuronal cell lines such as SH-SY5Y.[57] In experiments performed here, SH-SY5Y cells were differentiated by sequential treatments of 3 days using serum-free medium synthetic retinoid EC23, then a 3 day serum-free medium treatment with brain-derived neurotrophic factor (BDNF). Neurite length of NF200 positive cells was measured using ImageJ simple neurite tracer plugin. Laminin was used as a positive control since it has been previously described to be an optimal surface substrate for SH-SY5Y neuronal differentiation.[58] SH-SY5Y cells differentiated on laminin showed extensive and long neurite formation (Figure 6). SH-SY5Y cells differentiated on PEDOT:PSS:GOPS and PEDOT:PSS:DVS (3%) both showed formation of new neurites, showing that both are biocompatible and do not inhibit neuroregeneration. However, both laminin and PEDOT:PSS:DVS (3%) showed significantly longer neurite length when compared to that obtained on GOPS (Figure 6). This suggests that PEDOT:PSS cross-linked with 3% DVS increases the support for neuroregeneration and thus improves this biomaterial.

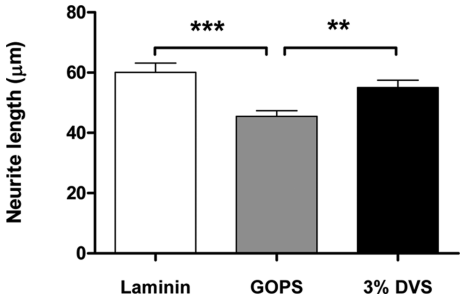


Figure 6. Neurite length analysis of differentiated SH-SY5Y cells cultured on PEDOT cross-linked with GOPS or 3% DVS. Statistical analysis performed with unpaired Student's *t* test with ** and *** denoting $p < 0.05$.

4. CONCLUSIONS

In this study we have presented DVS as an outstanding PEDOT:PSS cross-linker. It provided not only great potential for device fabrication due to the higher conductivity and transconductance in respect to GOPS but also a straightforward manner to create insoluble films at low temperature, making possible the incorporation of thermolabile and biological substances. The suspension has been characterized physicochemically by UV−vis−NIR, Raman, and IR spectroscopy. Conductivity measurements conducted on the dried material confirm the high applicability, giving a result of more than double conductivity, up to 600 S cm^{-1} (at 3% v/v) with respect to that with GOPS, and affirming that the addition of more cross-linker does not affect the conductivity. Biocompatibility assays were performed using L929 and SH-SY5Y cells, measuring cytotoxicity, proliferation, and neurite length, showing full biocompatibility and better support for neuroregeneration when compared to that with GOPS-cross-linked material. Furthermore, PEDOT:PSS:DVS (3%) showed a better initial cellular attachment of SH-SY5Y cells, which will be beneficial for a neural interface that has to be established. To show the applicability of PEDOT:PSS:DVS, the creation of an OECT and the measure of transconductance was tested. PEDOT:PSS:DVS channel values are high in the short and long term. This finding show its applicability in *in vivo* recording measurements. All-in-all, DVS cross-linking agent provides an outstanding solution for low-temperature fabrication of cross-linked PEDOT:PSS films for bioelectronics or other organic electronic applications.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acsami.7b02296.

Pictures of cross-linked and not-cross-linked film behavior in PBS solution, FT-IR spectra of PEDOT:PSS with different percentage of divinylsulfone, experimental procedure of polymerization of EDOT with PSS, NMR spectra of DVS-ethane diol mixture, film thickness measurements, OECTs outputs, frequency/transconductance dependence, and long-term OECT measurements (PDF)

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Notes

The authors declare no competing financial interest.

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